

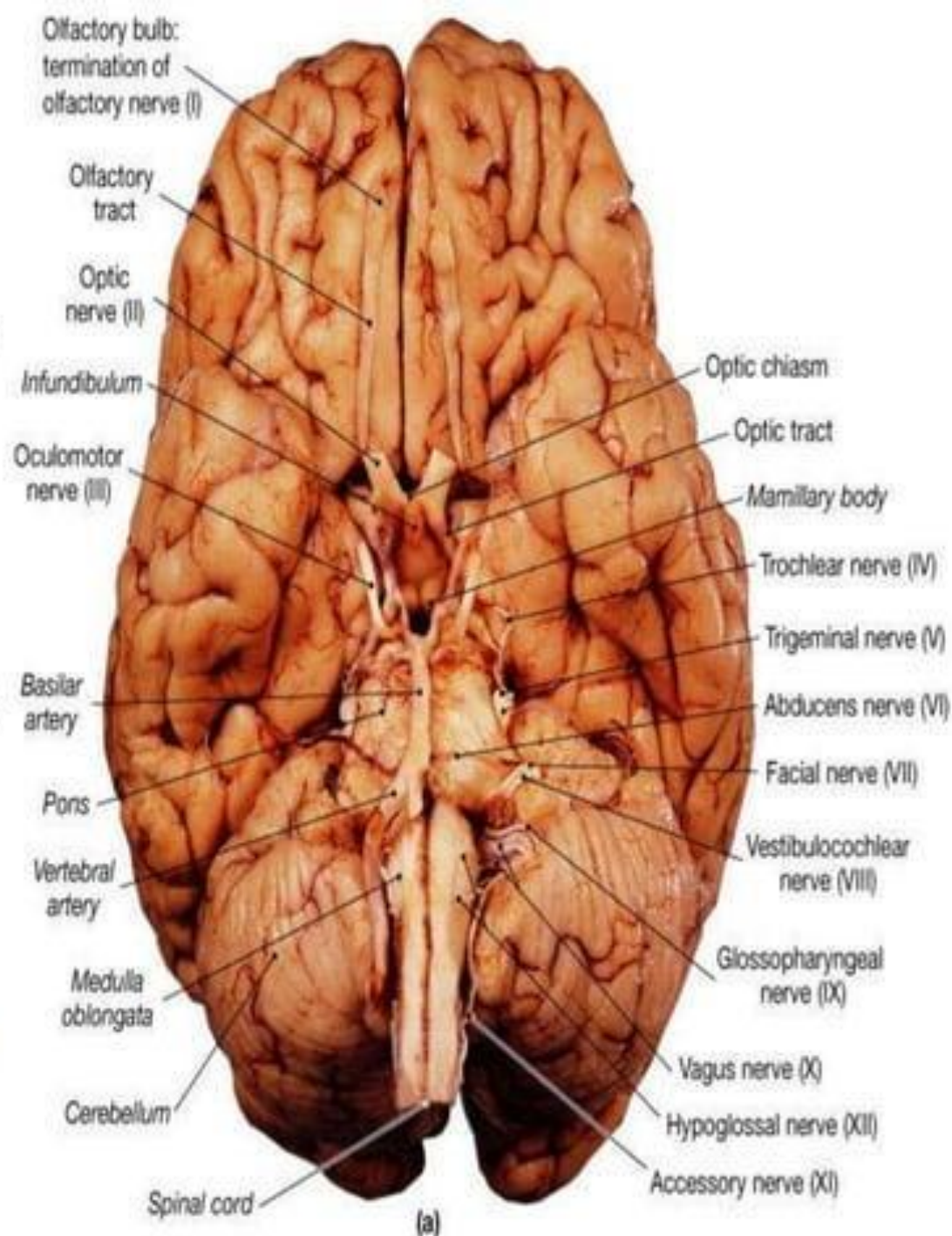
Overview Of Cranial Nerves

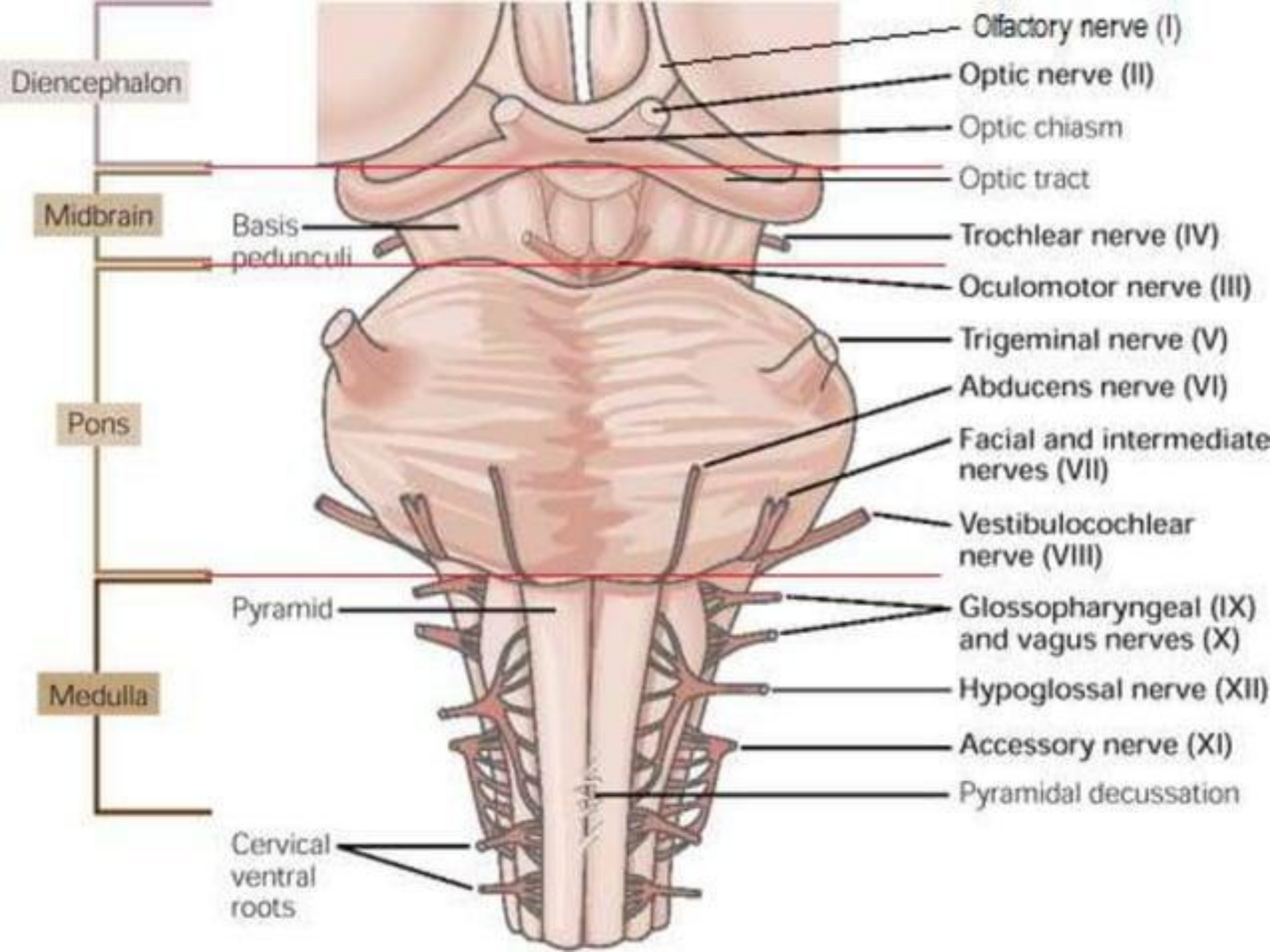
Dr Shahneela Siraj

DIKIOHS,DUHS.

INTRODUCTION

- There are **12 pairs** of cranial nerves that supply structures in the head, neck, thorax and abdomen.
- A cranial nerve can be made up of a mixture of functions which are called **modalities** or may be made up of a single modality.
- A modality is **sensory, motor, special sensory, etc.**





How to Remember CN I-XII

Oh! Oh! Oh!

To Touch And Feel

Very Good Velvet!

Ah Heaven! _____

Mnemonic for CN Function

- Some (CN I)
- Say (CN II)
- Marry (CN III)
- Money (CN IV)
- But (CN V)
- My (CN VI)
- Brother (CN VII)
- Says (CN VIII)
- Big (CN IX)
- Brains (CN X)
- Matter (CN XI)
- Most! (CN XII)

■ S = Sensory function

M = Motor function

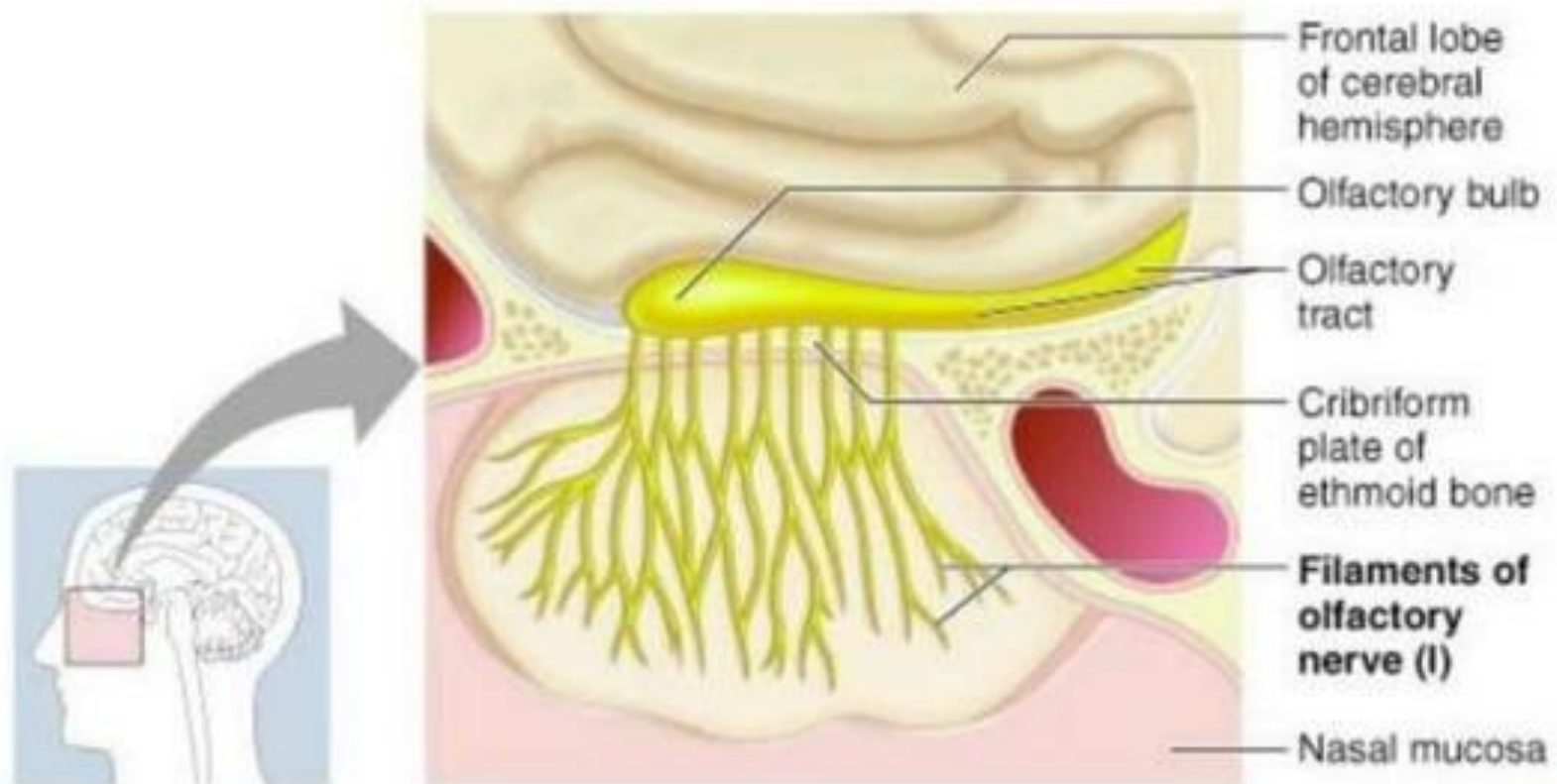
■ B = BOTH (Sensory and Motor function)

I.OLFACTORY NERVE

- It is **sensory** nerve
- Carries impulses for **sense of smell**
- **ORIGIN:** olfactory epithelium
- **OPENING IN SKULL:** opens in cribriform plate of ethmoid bone to receptors in roof of nasal cavity.
- Attaches to cerebrum.
- Lesion leads to **bilateral anosmia** can be caused by disease of olfactory mucous membrane, such as the common cold .



CRANIAL NERVE I: OLFACTORY

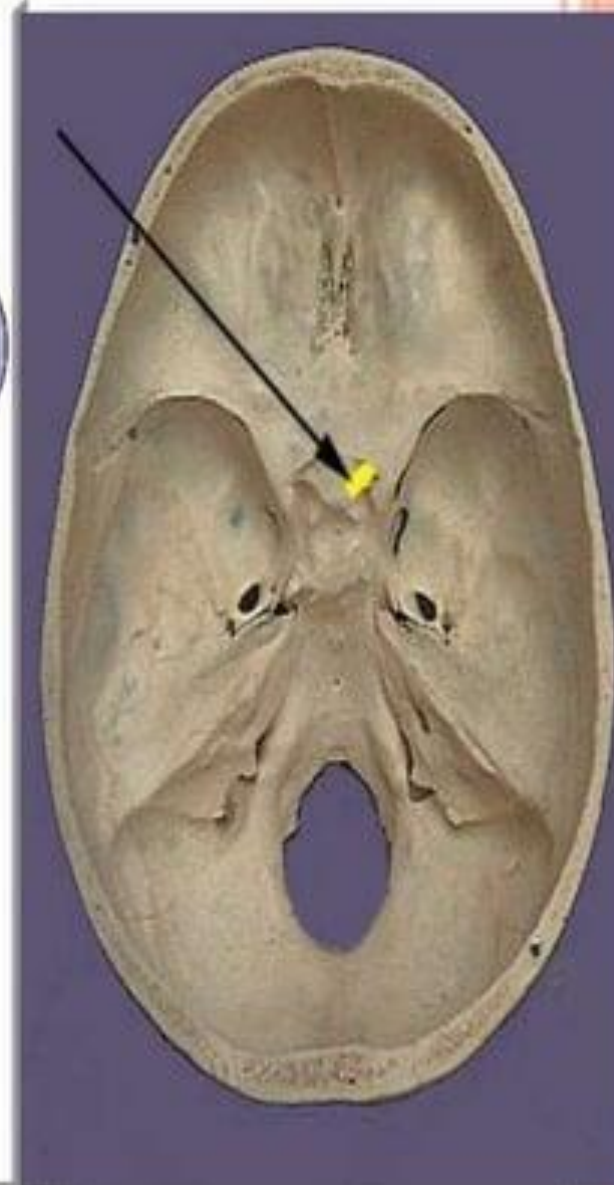
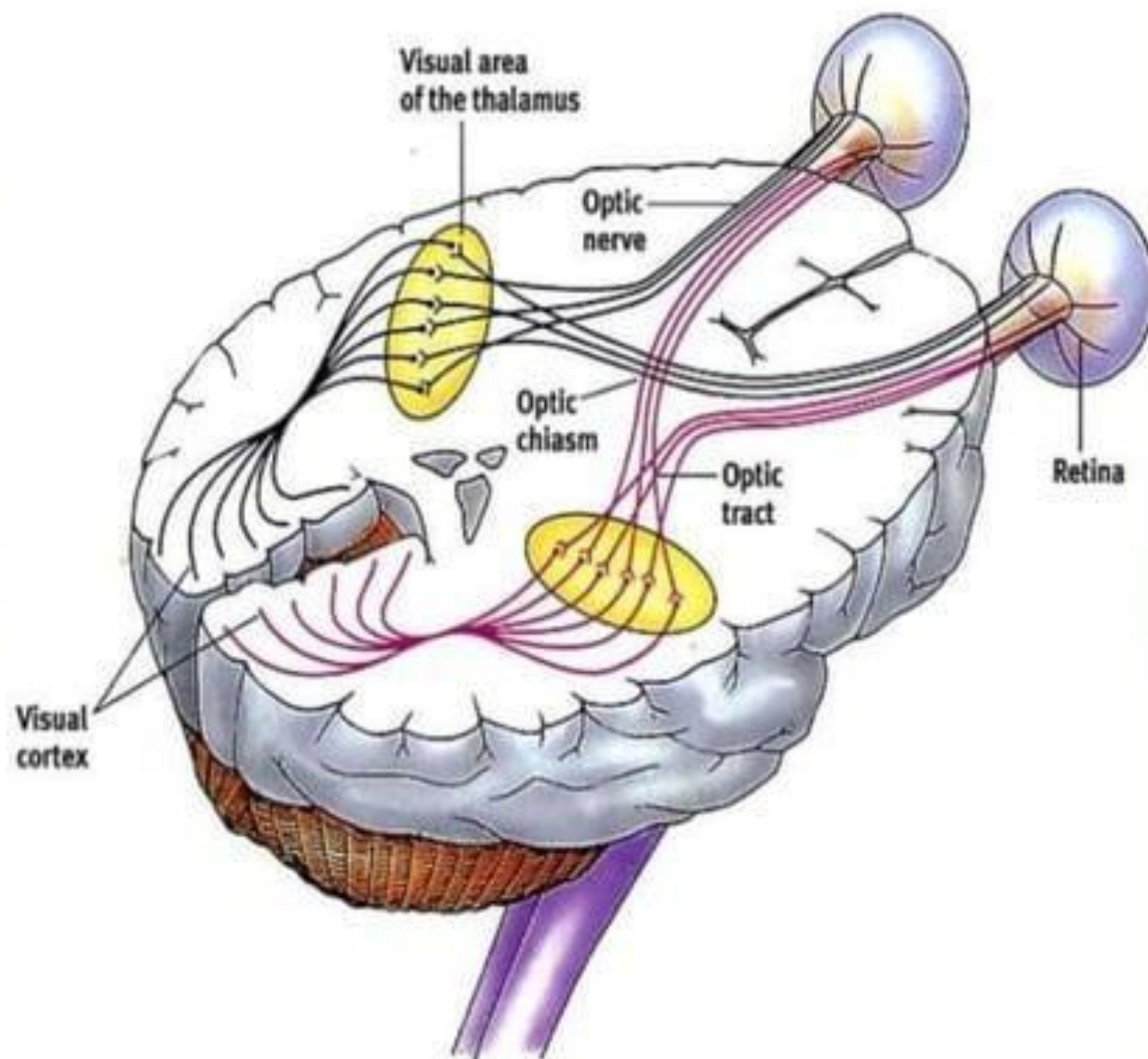


II.OPTIC NERVE

- It is **sensory** nerve.
- It carry **impulses for vision**.
- **Origin:** Back of eye ball/ retina of eye.
- **Opening to skull:** Optic canal and from there it converge to form optic chiasm.
- Attaches to diencephalon.
- Reflexes generated by this nerve are light reflex by lateral geniculate body, visual reflex & corneal reflex by blinking of eyes.
- Lesion leads to total blindness of one eye, hemianopia(partial lesion of optic chiasma on its lateral side.



II. OPTIC NERVE



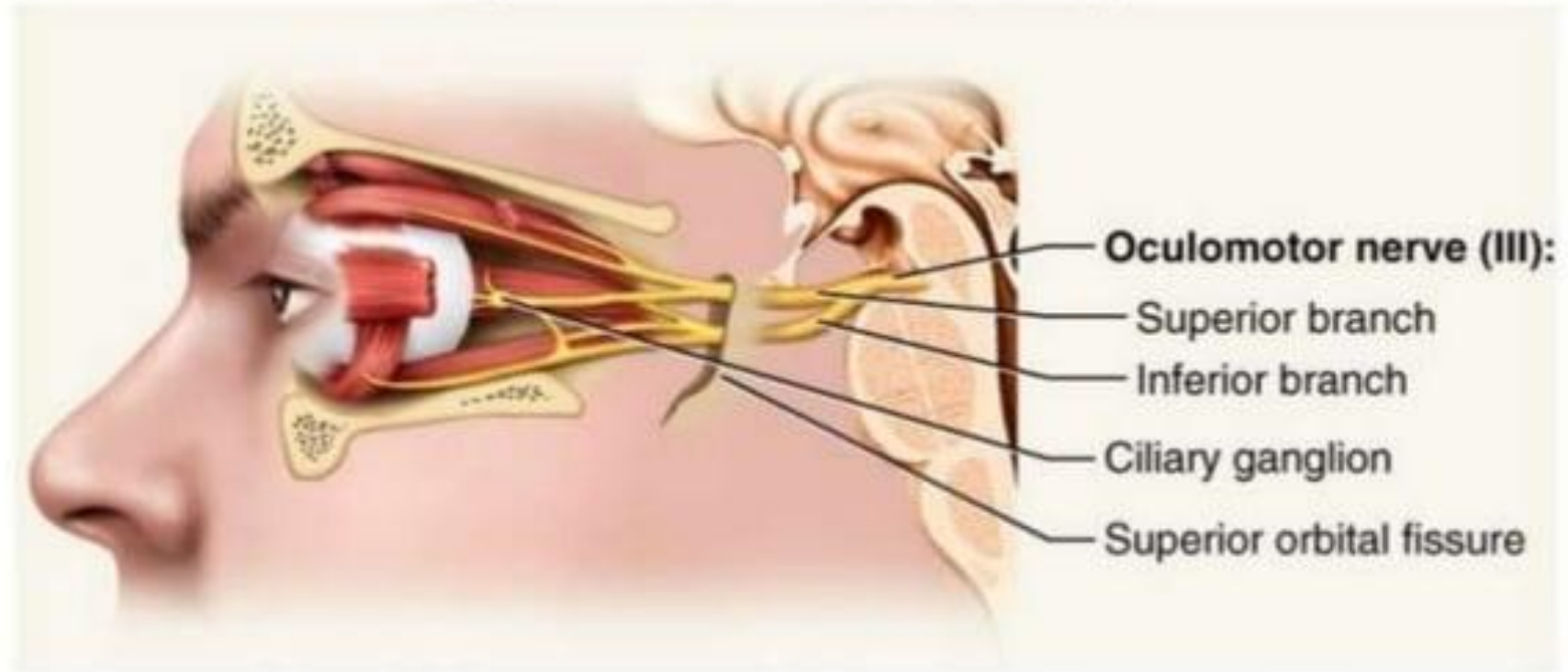
III. OCULOMOTOR NERVE



- It is **motor** nerve.
- **Function:**
 1. raises the upper eyelid.
 2. turn eye ball upward, downward & medially.
 3. constricts pupil.
 4. accommodates the eye
- The later **two functions** are parasympathetically controlled.
- Parasympathetic cell bodies are in **ciliary ganglia**.
- **Origin:** anterior surface of midbrain.
- **Opening in skull:** Superior orbital fissure.
- Lesion leads to **drooping of the upper eyelid (ptosis)** due to paralysis of levator palpebrae superioris muscle.
- Conditions effecting oculomotor nerve are diabetes, aneurysm, tumor, trauma, inflammation & vascular disease



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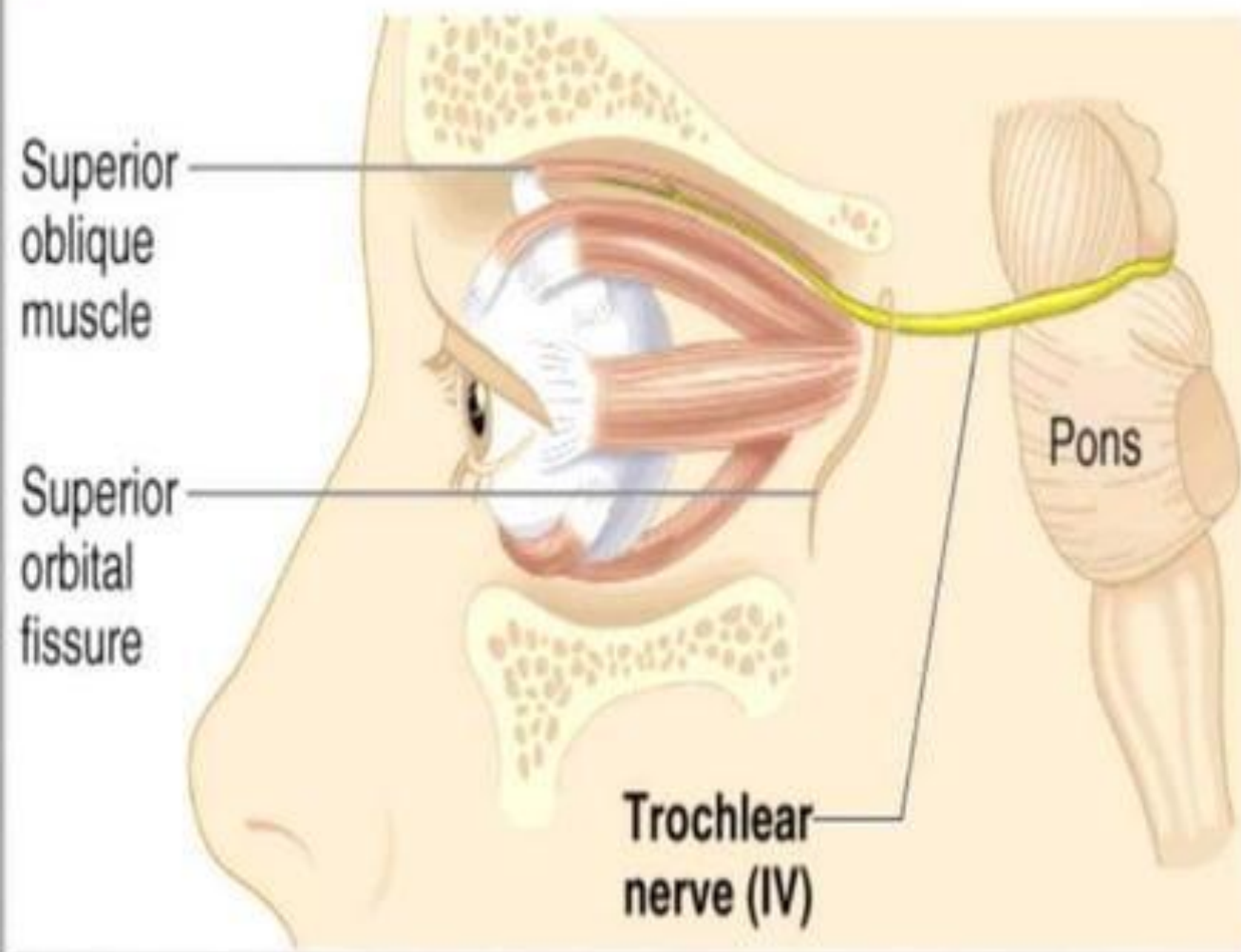


IV.TROCHLEAR NERVE

- It is motor nerve.
- **Function:** Assisting in turning eyeball downward and
- **Origin:** Posterior surface of the midbrain & innervate superior oblique muscle.
- **Opening to the Skull:** Superior orbital fissure.
- Attaches to midbrain.
- Lesion is due to aneurysm of internal carotid artery & vascular lesion of dorsal part of midbrain.
- Patient complains of double vision on looking downward.

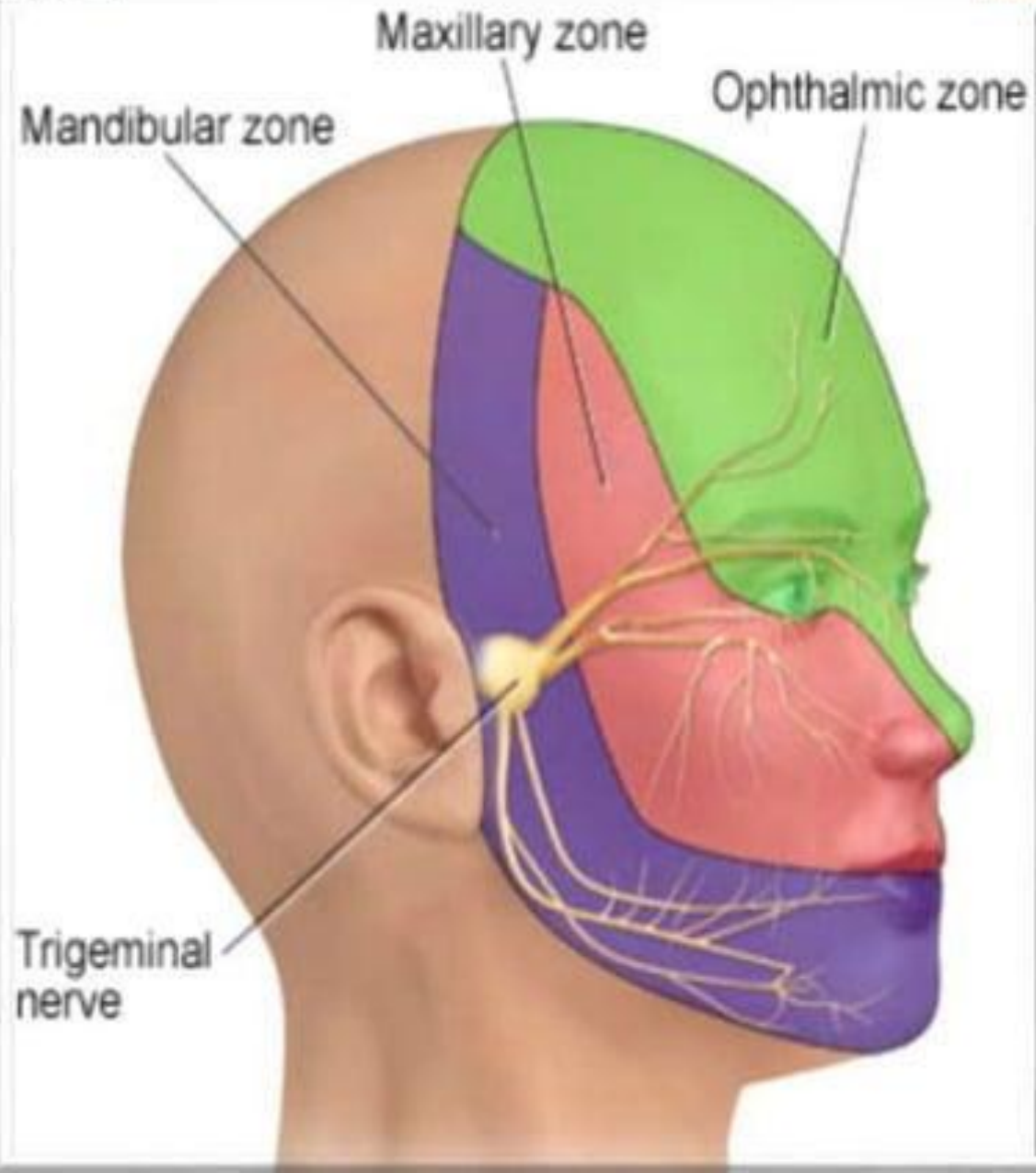


IV. TROCHLEAR NERVE



V. TRIGEMINAL NERVE

- It has **three** divisions as:
- Ophthalmic division
- Maxillary division
- Mandibular division

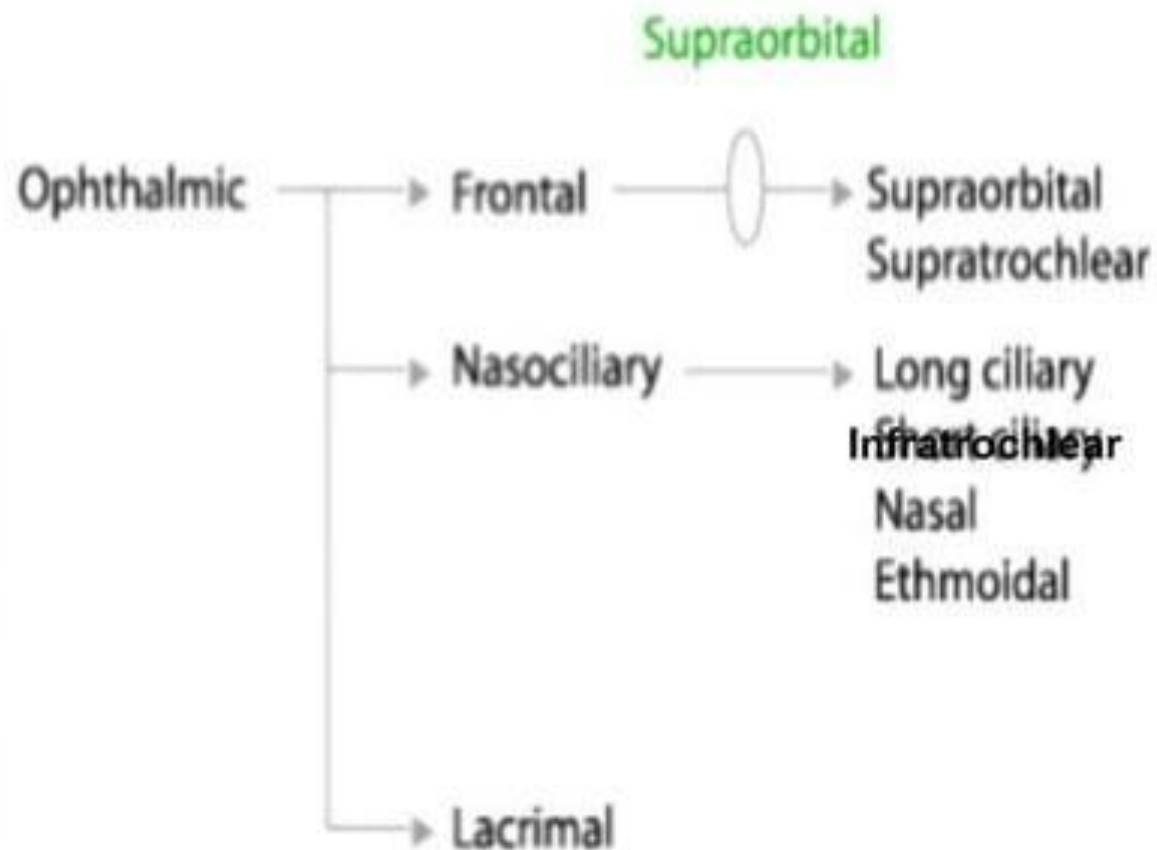


V1.OPHTHALMIC NERVE

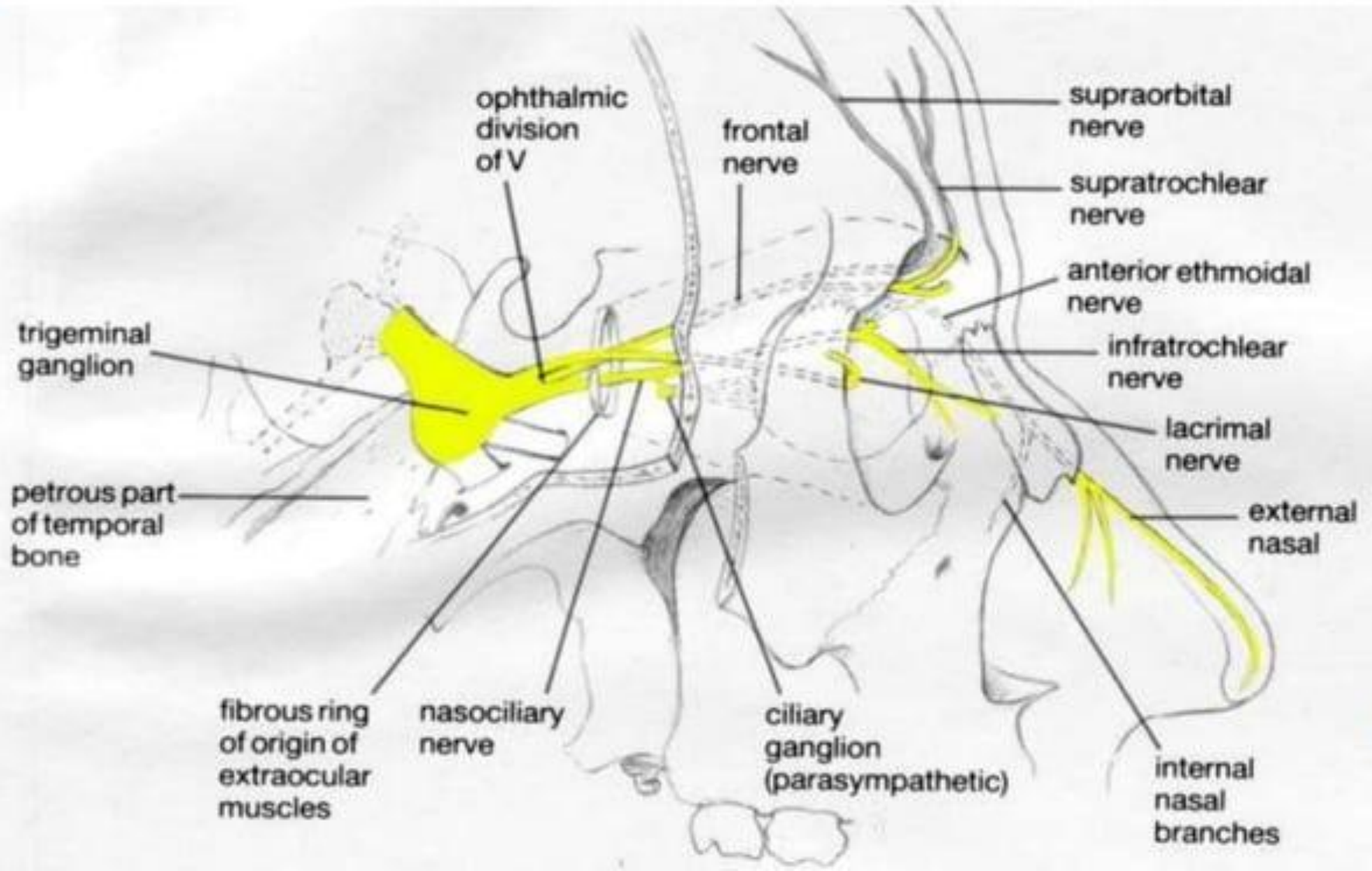
- It is sensory.
- **Function:**
cornea, skin of forehead, scalp, eyelids, nose, also mucous membrane of paranasal sinuses & nasal cavity.
- **Origin:** Anterior aspect of pons.
- **Opening in skull:** Superior orbital fissure.
- Exit orbit through supra orbital foramen.
- In lesion of this nerve cornea & conjunctiva will be insensitive to touch.



V1. OPHTHALMIC NERVE



V1. Ophthalmic Nerve



V2.MAXILLARY NERVE

- It is sensory nerve.

- **Function:**

skin of face over maxilla.

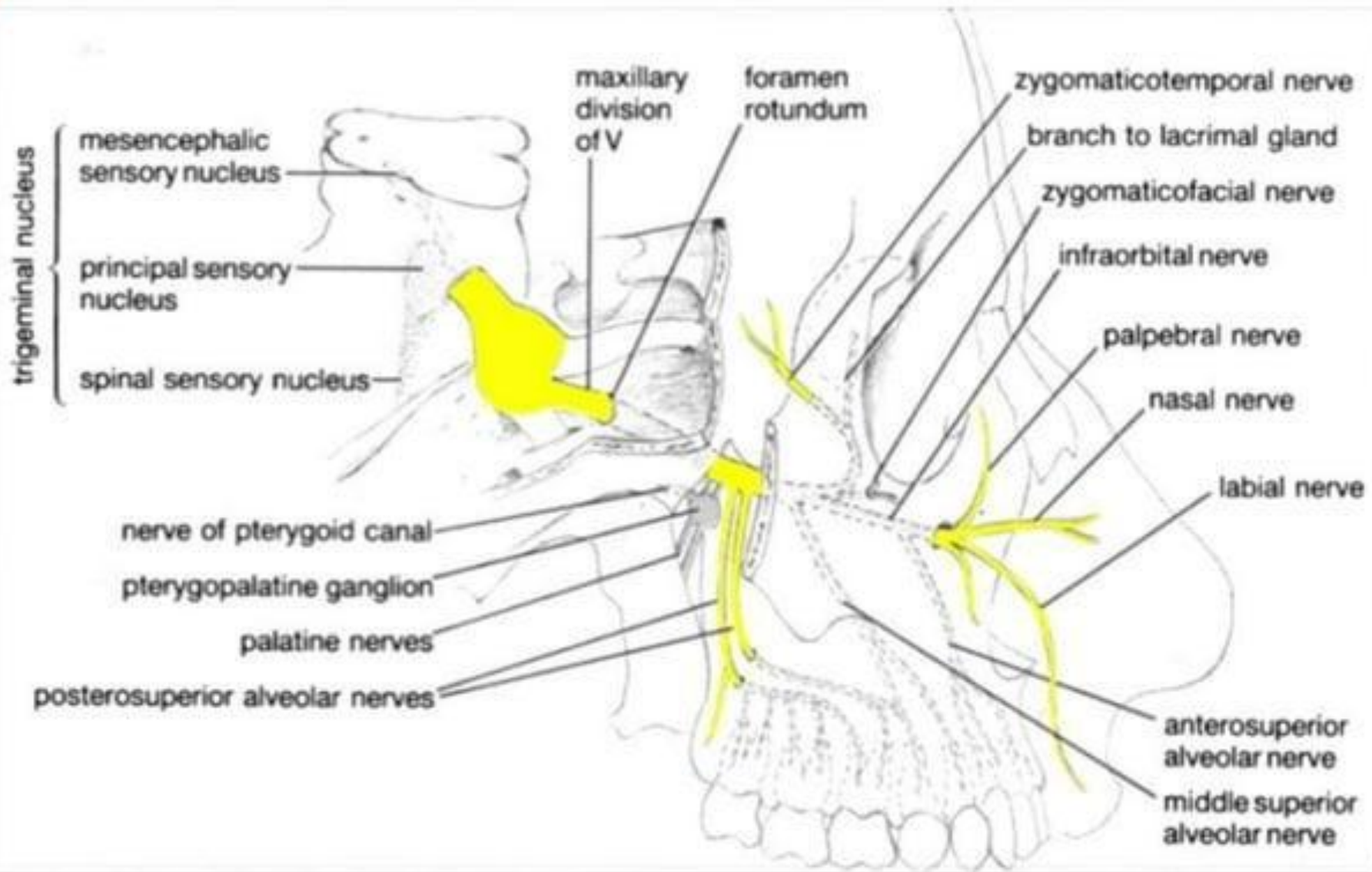
Teeth of upper jaw.

Mucous membrane of nose , the maxillary sinus & palate.

- **Origin:** Anterior aspect of pons.
- **Opening in skull:** Foramen rotundum
- Exit through infraorbital foramen.



V2. Maxillary Nerve



V3. MANDIBULAR NERVE

- **Component:** a. Motor

- **Function:**
 - Muscles of mastication
 - Mylohyoid
 - Anterior belly of digastric
 - Tensor veli palatine
 - Tensor tympani

- **Origin:** Anterior aspect of the pons

- **Opening to the Skull:** Foramen ovale
- 

V3. MANDIBULAR NERVE

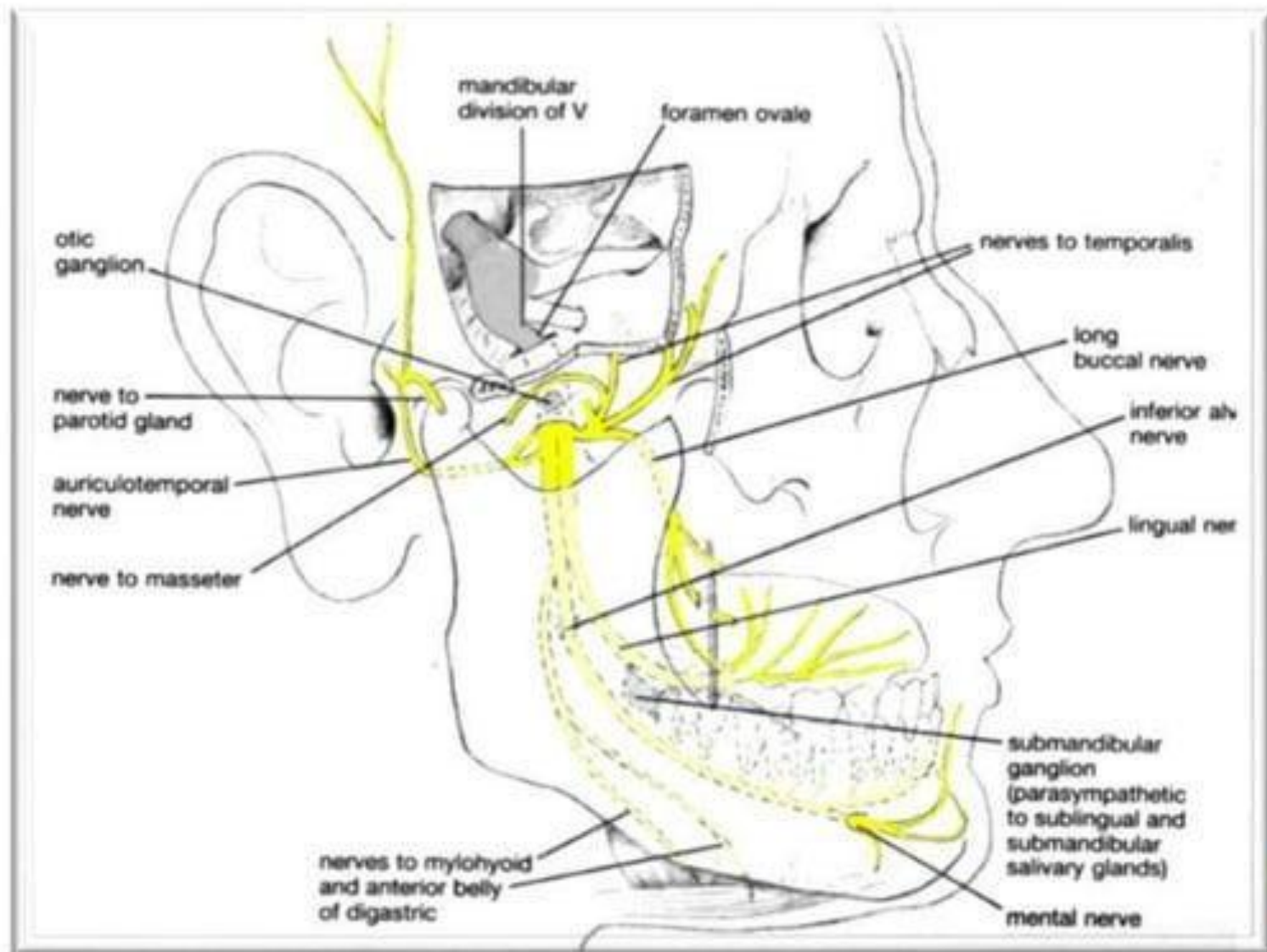
- **Component:** b. Sensory

- **Function:**
 - Skin of cheek
 - Skin over mandible and side of head
 - Teeth of lower jaw and TMJ
 - Mucous membrane of mouth and anterior part of tongue

- **Origin:** Anterior aspect of the pons

- **Opening to the Skull:** Foramen ovale

V3. Mandibular Nerve

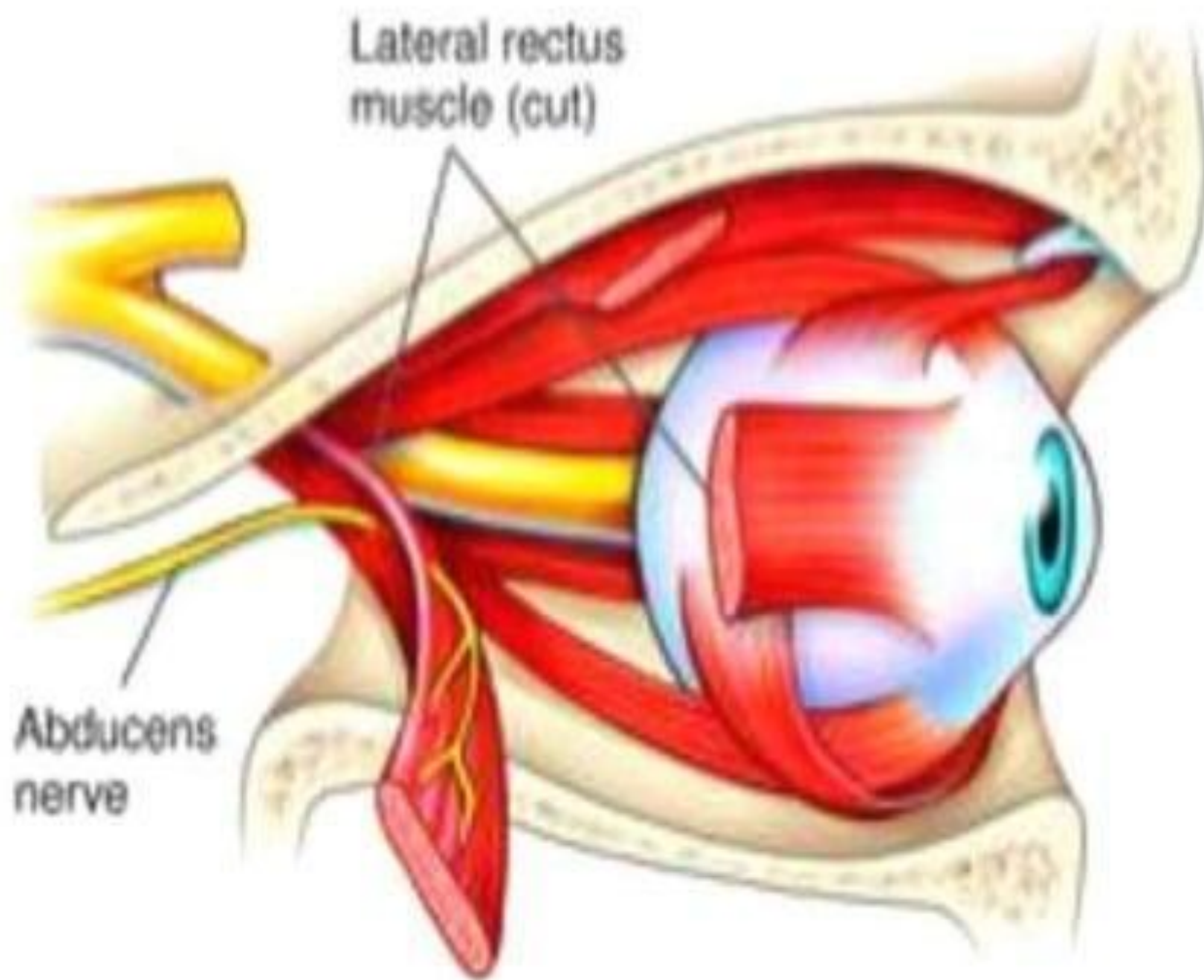


VI.ABDUCENT NERVE

- It is motor nerve
- **Function:** Lateral rectus muscle turns eyeball laterally
- **Origin:** Anterior Surface of hindbrain between pons and medulla
- **Opening to the Skull:** Superior orbital fissure
- Fibers leaves the inferior pons & enter orbit via superior orbital fissure.
- Patient can't turn the eye laterally.
- Lesions include damage due to head injuries, cavernous sinus thrombosis or aneurysm of internal carotid artery & vascular lesions of pons.



VI. Abducent Nerve



VII.FACIAL NERVE

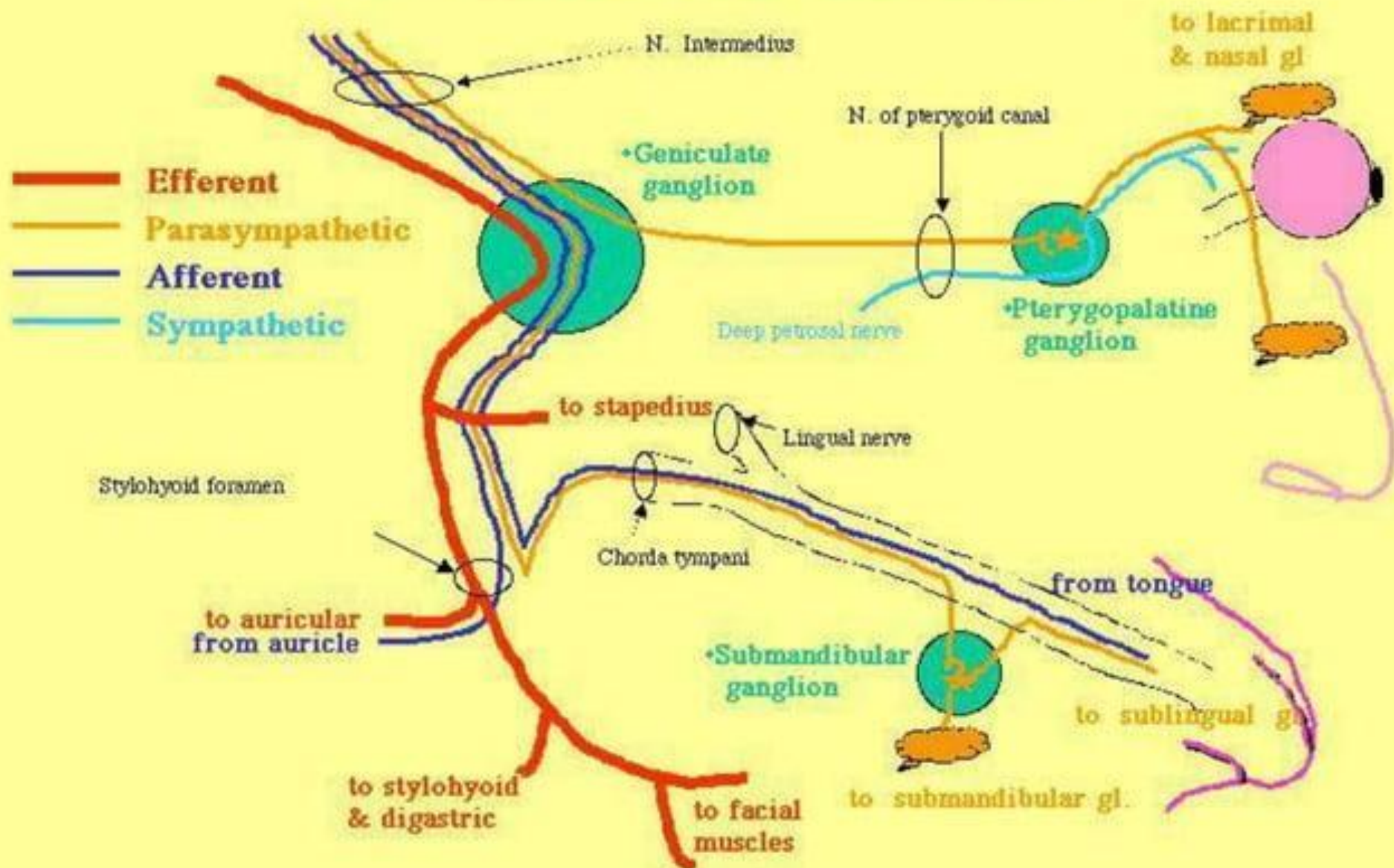
- It is mixed nerve.
 - **Function:-**
 - **Motor:** Muscles of face & scalp, stapedius muscle, posterior belly of digastric & stylohyoid muscle.
 - **Sensory:** Taste from anterior 2/3rd of tongue, from floor of mouth & palate.
 - **Secretomotor parasympathetic:** submandibular & sublingual salivary glands, the lacrimal gland & glands of nose & palate.
 - **Opening in the skull:** internal acoustic meatus, facial & stylomastoid foramen.
 - Attaches to pons.
 - **Effects of damage:** inability to control facial muscles; distorted sense of taste.
- 

BELL'S PALSY

- Paralysis of facial muscles of affected side
- Loss of taste sensation
- Caused by herpes simplex virus.
- Lower eyelids droops.
- Corner of mouth sags.
- Tears drips continuously & eye cannot be completely closed.
- Condition may disappear spontaneously without treatment.



Facial Nerve



VII. VESTIBULOCOCHLEAR

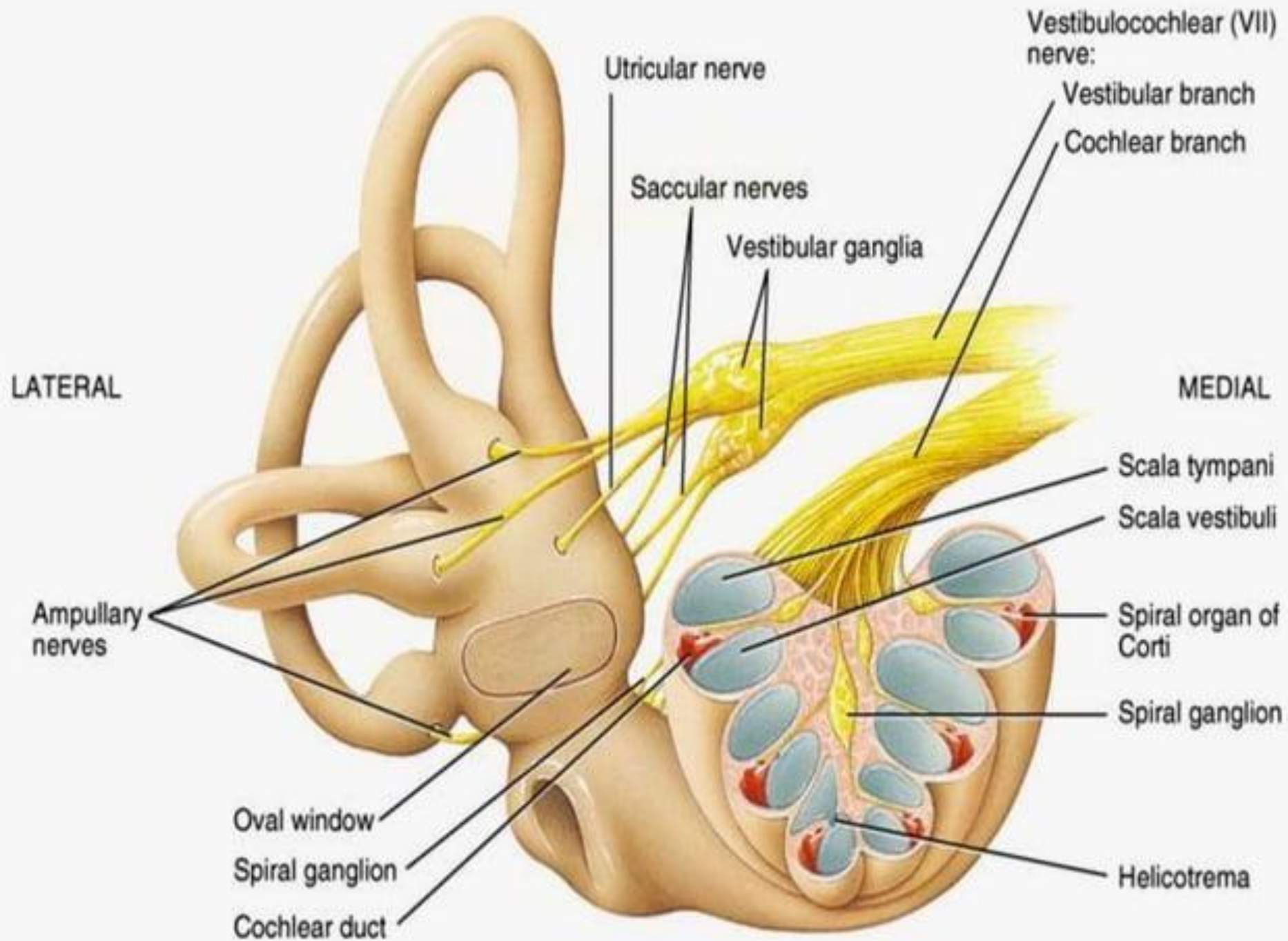
- **VESTIBULAR NERVE:-**

- It is sensory nerve.
- **Function:** Position & movement of head.
- **Opening in the skull:** Internal acoustic meatus

- **COCHLEAR NERVE:-**

- It is sensory nerve.
- **Function:** Organ of corti_ hearing.
- **Opening in the skull:** internal acoustic meatus.

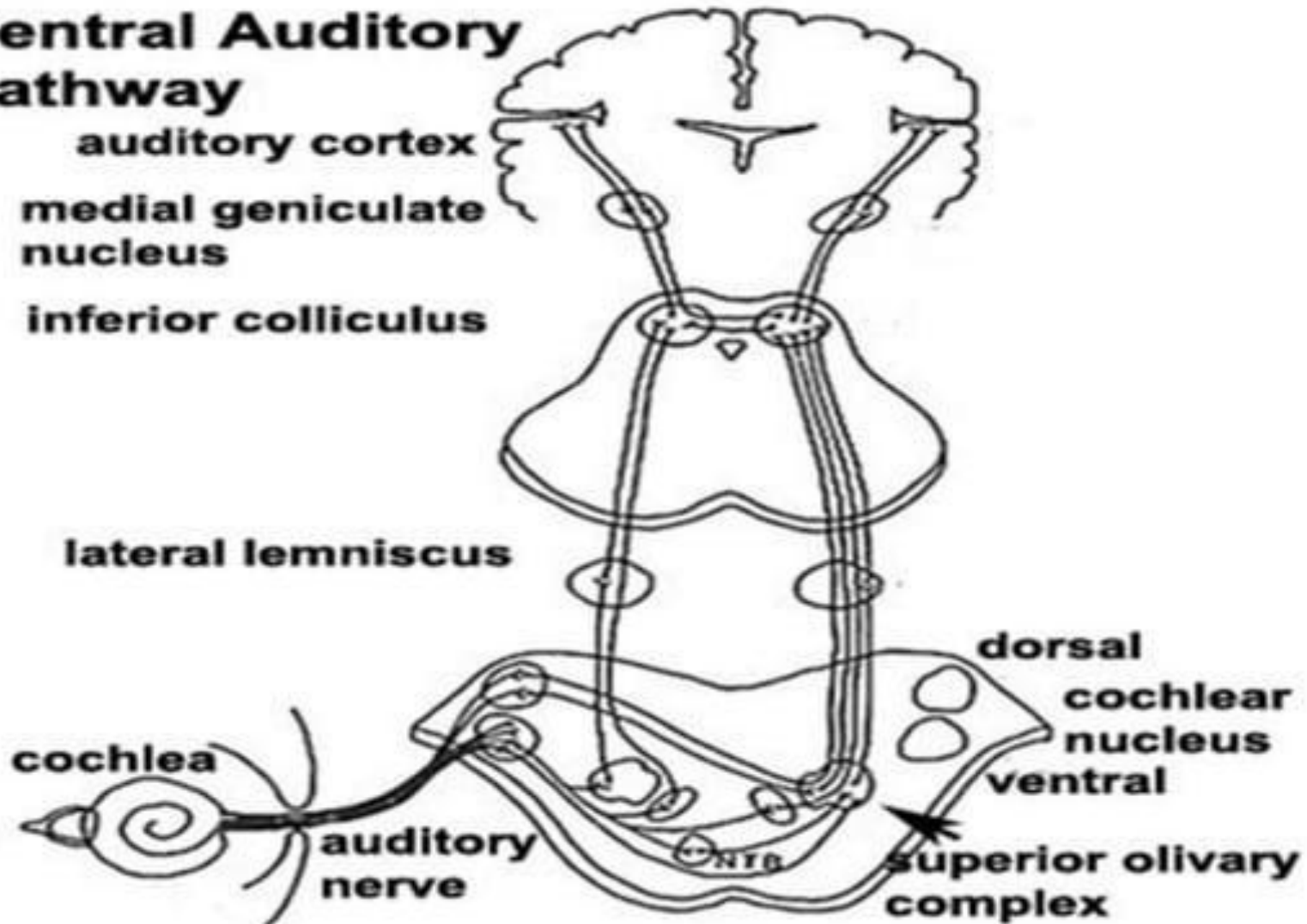




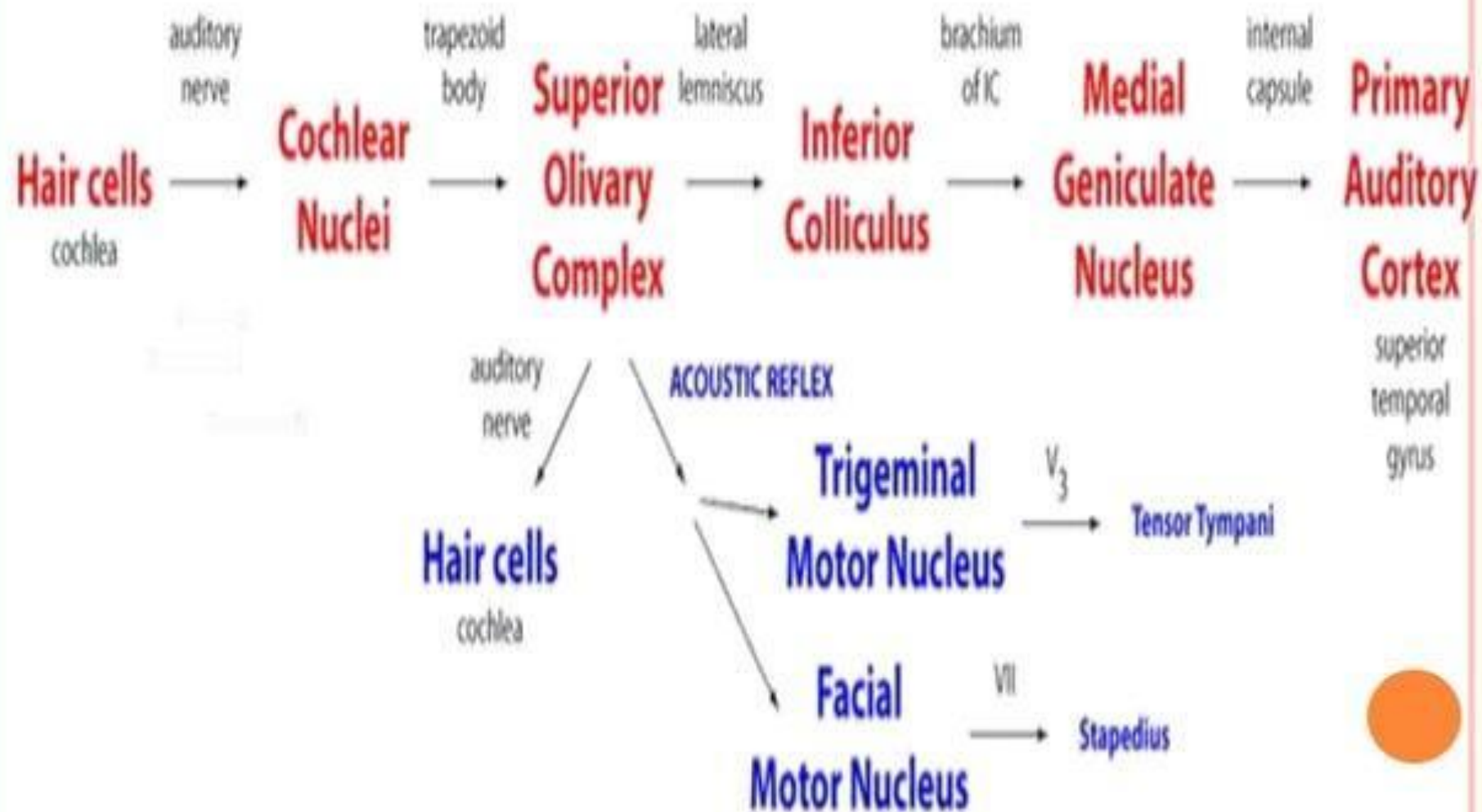
AUDITORY PATHWAY

Central Auditory Pathway

auditory cortex
medial geniculate nucleus
inferior colliculus



AUDITORY PATHWAY



SYMPTOMS

- Damage to the vestibulocochlear nerve may cause the following symptoms:
- hearing loss
- vertigo
- false sense of motion
- loss of equilibrium (in dark places)
- nystagmus
- motion sickness
- gaze-evoked tinnitus.



PATHOLOGIES

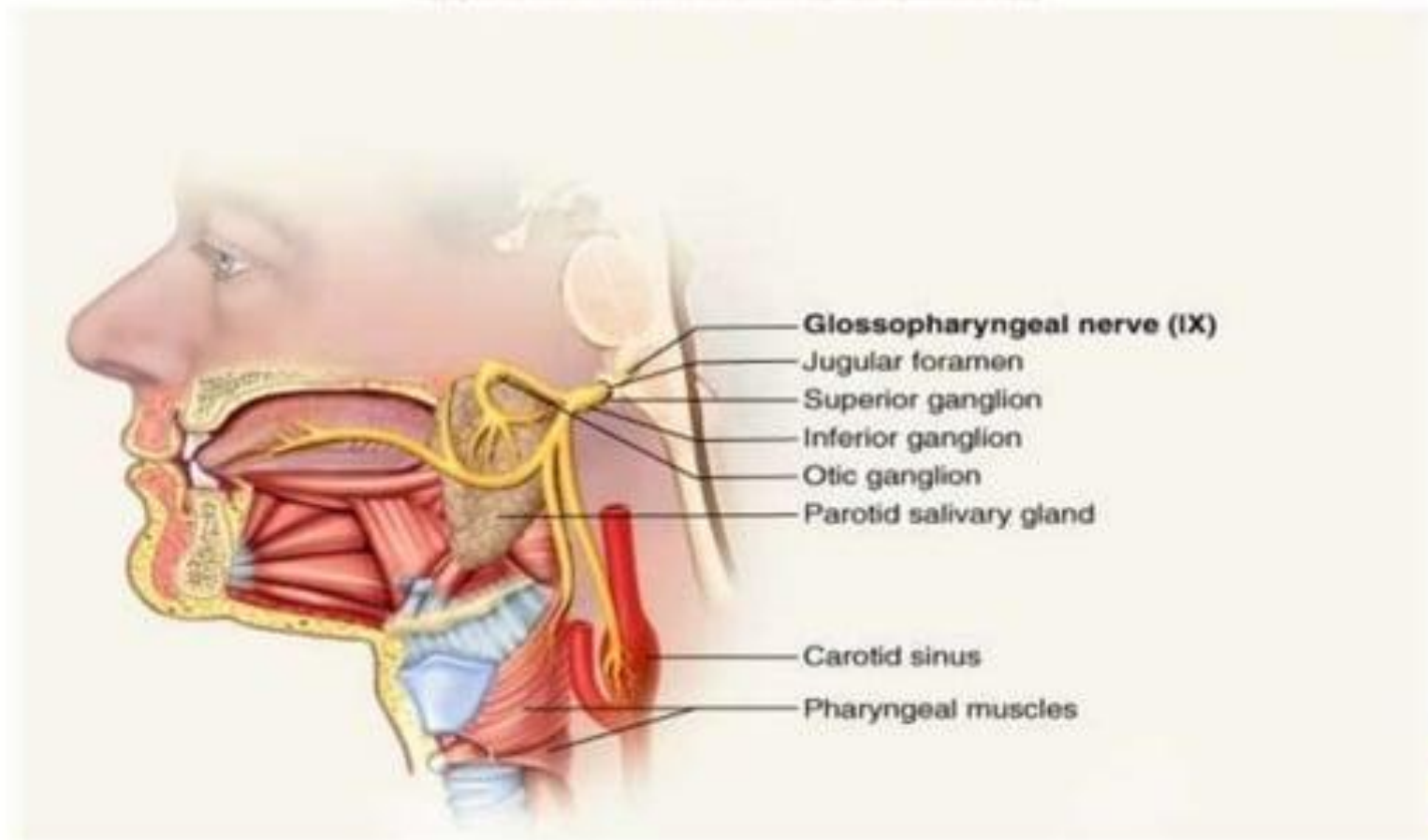
- Hearing loss
- Central auditory processing disorders (**CAPD**)
- Tinnitus
- Presbycusis
- Nystagmus



XI.GLOSSOPHARYNGEAL

- It is mixed nerve.
 - **Function:-**
 - **Motor:** Stylopharyngeus muscle assists swallowing.
 - **Secretomotor parasympathetic:** parotid salivary gland.
 - **Sensory:** general sensation & taste from posterior 1/3 of tongue & pharynx, carotid sinus & carotid body(chemoreceptor).
 - **Opening in the skull:** Jugular foramen.
 - It attaches to medulla oblongata.
 - **Clinical tests:** gag reflex, swallowing, and coughing
 - **Effects of damage:** difficulty swallowing
- 

CRANIAL NERVE IX: GLOSSOPHARYNGEAL



X.VAGUS

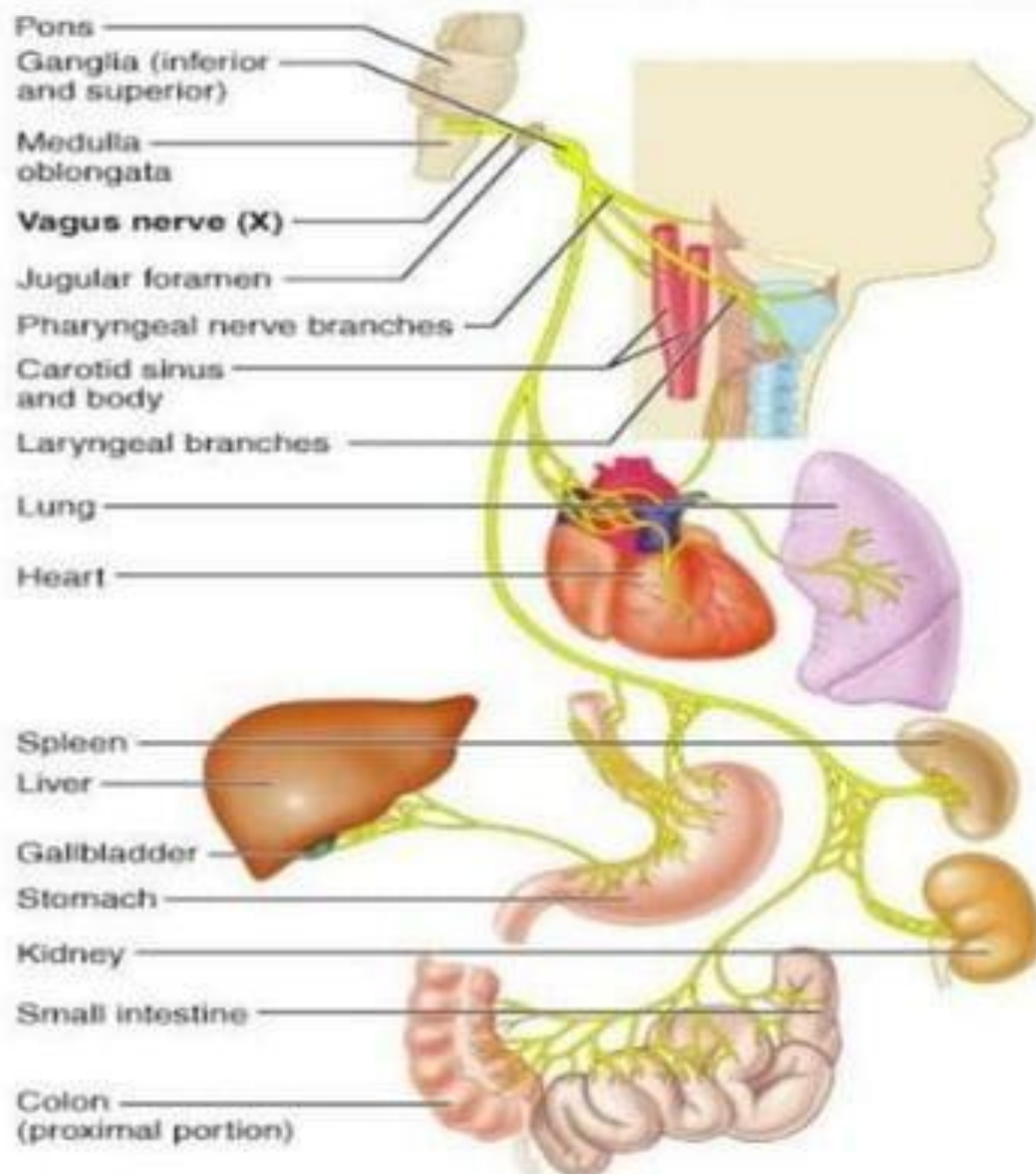
- It is mixed nerve.
- **Nick name:** Wanderer.
- **Function:-**
- **Motor & sensory:** Heart & great thoracic blood vessels , larynx, trachea, bronchi & lungs, alimentary tract from pharynx to splenic flexure of colon, liver, kidneys & pancreas.
- **Opening in the skull:** Jugular foramen.
- Effects of damage: hoarseness or loss of voice; dysphagia, cardiovascular problems, digestive problems, urinary incontinence, deafness, palatal function, gag reflex, spastic dysarthria.
- **Gag Reflex:**
- A normal reflex consisting of elevation of the palate, retraction of the tongue, and contraction of the throat muscles.

BRANCHES OF VAGUS NERVE

- Auricular nerve
- Pharyngeal nerve
- Superior laryngeal nerve
- Superior cervical cardiac branches of vagus nerve
- Inferior cervical cardiac branch
- Recurrent laryngeal nerve
- Thoracic cardiac branches
- Branches to the pulmonary plexus
- Branches to the esophageal plexus
- Anterior vagal trunk
- Posterior vagal trunk



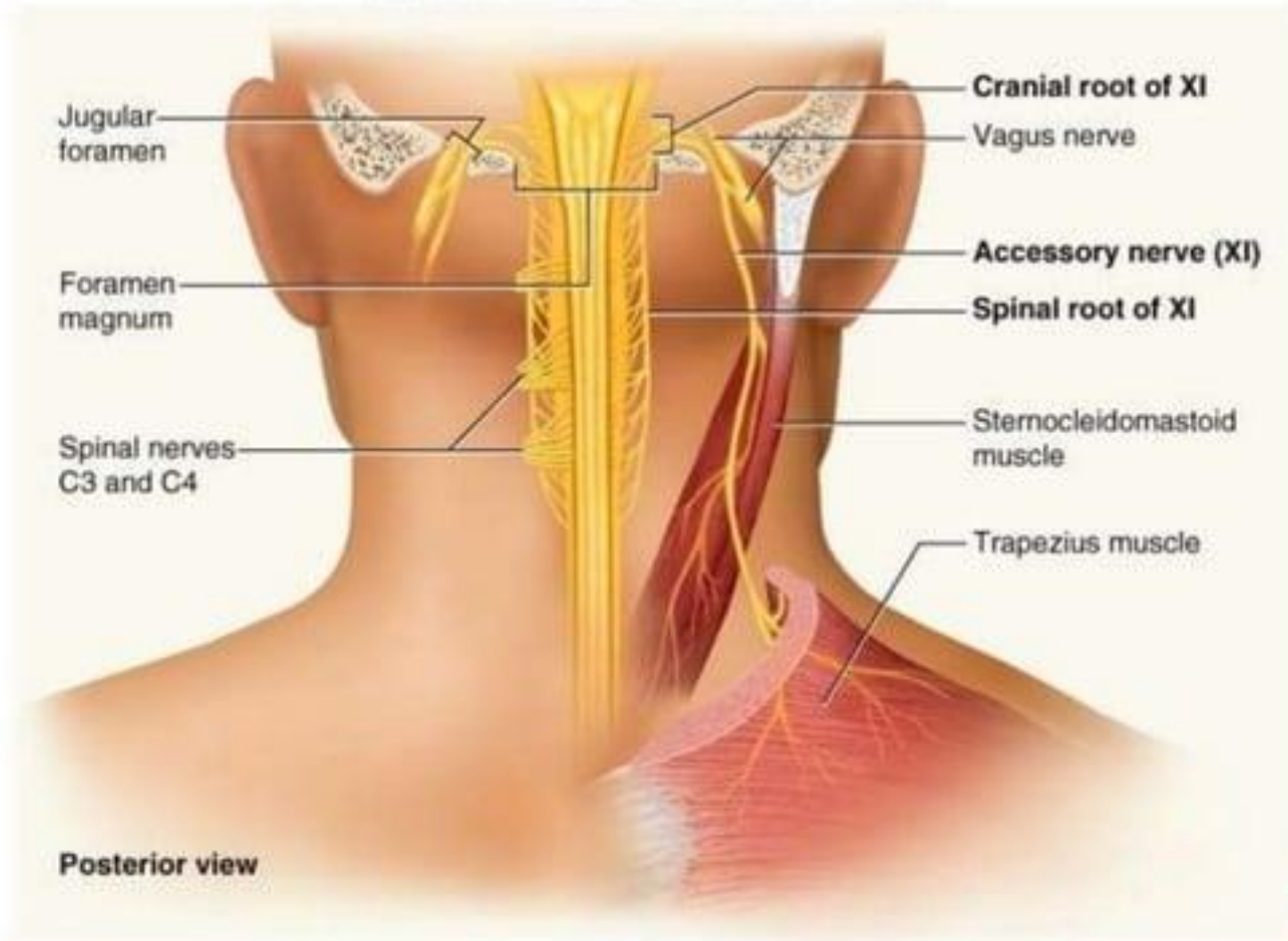
CRANIAL NERVE X: VAGUS



XI.ACCESSORY

- It is motor nerve.
- Formed from **cranial root** emerging from medulla & a **spinal root** arising from the superior region of spinal cord.
- **Function:** swallowing; head, neck, and shoulder movements.
- **Opening in the skull:** Jugular foramen.
- Spinal root passes upward into cranium via the foramen magnum.
- Accessory nerve leaves the foramen via the jugular foramen.
- **Clinical tests:** rotate head and shrug shoulders against resistance
- **Effects of damage:** impaired movement of head, neck, and shoulders; paralysis of sternocleidomastoid

CRANIAL NERVE XI: ACCESSORY



XII.HYPOGLOSSAL

- It is motor nerve.
 - **Function:-** Muscles of tongue (except palatoglossus) controlling its shape & movement.
 - **Opening in the skull:** Hypoglossal canal.
 - Innervates both extrinsic & intrinsic muscles of tongue.
 - **Clinical test:** tongue function
 - **Effects of damage:** difficulty in speech and swallowing; atrophy of tongue; inability to stick out (protrude) tongue
 - Lesions occur from demyelinating diseases & vascular accidents.
- 

CRANIAL NERVE XII: HYPOGLOSSAL

Hypoglossal canal, hypoglossal nerve, and hypoglossal muscle

